

STA2023 – Chapter 4: Probability — In-Class Notes, FILLED IN

(Blanks from the notes are filled with the textbook (Sullivan) definitions; every example is worked with all steps. Round probabilities to 4 decimals.)

4.1 Probability Rules

Random process = a situation where the outcome of a single trial cannot be predicted with certainty, but a long-run pattern (proportion) appears after many trials.

Coin flip example: after 100 flips the proportion of heads settles just above 0.50 → the chance of a head on this coin is about **0.5** (the long-run proportion). This is the **Law of Large Numbers**: as the number of repetitions grows, the proportion of times an outcome occurs gets closer to its probability.

Probability = a measure of the likelihood of a random event; the long-term proportion of times an outcome is observed if the experiment is repeated many times.

Experiment = any process with uncertain results that can be repeated. *Examples*: flipping a coin, rolling a die, drawing a card, picking an M&M from a bag.

Sample space, S = the collection of ALL possible outcomes of an experiment. *Example*: rolling a die → $S = \{1, 2, 3, 4, 5, 6\}$.

Event = any collection (subset) of outcomes from the sample space that we are interested in; written with capital letters (E, F, ...). A **simple event** has exactly one outcome.

Example 1 (roll one fair six-sided die) - (a) Outcomes: 1, 2, 3, 4, 5, 6 - (b) $S = \{1, 2, 3, 4, 5, 6\}$ - (c) $E = \text{"roll an even number"} = \{2, 4, 6\}$

Example 2 (one M&M picked at random) - (a) Outcomes: brown, yellow, red, blue, orange, green - (b) $S = \{\text{brown, yellow, red, blue, orange, green}\}$ - (c) $E = \text{"red or blue"} = \{\text{red, blue}\}$

Basic Rules of Probability

Notation: $P(E)$ = the probability that event E occurs. $N(E)$ = number of outcomes in E; $N(S)$ = number of outcomes in the sample space.

1. The probability of any event is between 0 and 1: **$0 \leq P(E) \leq 1$** . ($P(E) = 0$ → impossible; $P(E) = 1$ → certain.)
2. The probabilities of all outcomes in the sample space add to 1: **$P(e_1) + P(e_2) + \dots + P(e_n) = 1$** .

Probability model = a table listing every possible outcome and its probability (must satisfy rules 1 and 2).

Unusual event = an event with a low probability, usually **less than 0.05 (5%)**. (A 3% chance → unusual.)

Example 3 (die model, each side 1/6): every probability is between 0 and 1 ✓, and $1/6 \times 6 = 6/6 = 1$ ✓ → **valid**.

Example 4 - Model 1 (0.50, 0.50, 0): all between 0 and 1, sum = 1 → **valid** (a probability of 0 is allowed). - Model 2 (0.40, 0.20): sum = 0.60 $\neq 1$ → **not valid**. - Model 3 (−0.10, 0.40, 0.60, 0.10): −0.10 is negative → **not valid** (even though the sum is 1).

Methods to Determine Probabilities

Empirical method — uses relative frequency from data already collected: $P(E) \approx (\text{frequency of } E) \div (\text{number of trials})$

Example 6 (200 people, means of travel to work) (a) Probability model = relative frequencies (frequency $\div$ 200):

Means of travel	Frequency	Probability
Drive alone	83	$83/200 = 0.415$
Carpool	22	$22/200 = 0.110$
Public transportation	10	$10/200 = 0.050$
Walk	5	$5/200 = 0.025$
Other means	3	$3/200 = 0.015$
Work at home	77	$77/200 = 0.385$
Total	200	1.000 ✓

(b) $P(\text{carpool}) = 22/200 = \mathbf{0.11}$. Interpretation: if we randomly selected many individuals, about 11% of them (about 11 out of every 100) would carpool to work. (c) $P(\text{walk}) = 5/200 = 0.025 < 0.05 \rightarrow$ **yes, unusual**.

Example 7 (M&M colors, total = $13 + 14 + 13 + 24 + 20 + 16 = 100$) (a) Brown $13/100 = 0.13$; Yellow 0.14; Red 0.13; Blue 0.24; Orange 0.20; Green 0.16 (sum = 1.00 ✓) (b) $P(\text{red}) = 13/100 = \mathbf{0.13}$

Classical method — requires **equally likely outcomes**; count instead of experiment: $P(E) = N(E) \div N(S) = (\text{number of ways } E \text{ can happen}) \div (\text{total number of possible outcomes})$

Example 8 (fair coin flipped twice) - (a) $S = \{HH, HT, TH, TT\} \rightarrow N(S) = 4$ - (b) $P(\text{two heads}) = P(HH) = \mathbf{1/4 = 0.25}$ - (c) "At least 1 tails" = $\{HT, TH, TT\} \rightarrow \mathbf{3/4 = 0.75}$ (or $1 - P(HH) = 1 - 1/4 = 3/4$)

Example 9 (pick 2 of Adam, Ben, Sam, Dustin) - (a) $S = \{AB, AS, AD, BS, BD, SD\} \rightarrow N(S) = 6$ (order doesn't matter) - (b) $P(\text{Adam and Ben}) = \mathbf{1/6 \approx 0.1667}$ - (c) $P(\text{Ben goes}) = \{AB, BS, BD\} \rightarrow \mathbf{3/6 = 1/2 = 0.5}$

Subjective method — a probability based on personal judgment, experience, or opinion (an educated guess), used when an experiment cannot be repeated and no data exist (e.g., "chance of a recession next year").

4.2 The Addition Rule and Complements

Disjoint (mutually exclusive) events = two events that have NO outcomes in common; they cannot both happen at the same time. In a Venn diagram the circles do not overlap.

Example 10 (chips 0–9; $E = "\leq 2" = \{0, 1, 2\}$; $F = "\geq 8" = \{8, 9\}$): no outcome in common $\rightarrow$ **E and F are disjoint**. $P(E) = 3/10$, $P(F) = 2/10$.

Addition Rule for Disjoint Events

If E and F are disjoint: $P(E \text{ or } F) = P(E) + P(F)$ (extends to any number of disjoint events: $P(E \text{ or } F \text{ or } G) = P(E) + P(F) + P(G)$). (Example 10: $P(E \text{ or } F) = 3/10 + 2/10 = 5/10 = 0.5$.)

Example 11 (one card from a 52-card deck) - (a) $P(\text{king}) = 4/52 = \mathbf{1/13 \approx 0.0769}$ - (b) $P(\text{king or queen or jack}) = 4/52 + 4/52 + 4/52 = 12/52 = \mathbf{3/13 \approx 0.2308}$ (disjoint: a card can't be two ranks) - (c) $P(\text{heart or diamond}) = 13/52 + 13/52 = 26/52 = \mathbf{1/2 = 0.5}$

The General Addition Rule

Chips 0–9; $E = \text{"odd"} = \{1, 3, 5, 7, 9\}$; $F = \text{"} \leq 4 \text{"} = \{0, 1, 2, 3, 4\}$. Outcomes 1 and 3 are in BOTH, so adding $P(E) + P(F)$ would count them twice. Count directly: $E \text{ or } F = \{0, 1, 2, 3, 4, 5, 7, 9\} \rightarrow 8/10$. Check: $5/10 + 5/10 - 2/10 = 8/10 \checkmark$.

General Addition Rule — for ANY two events E and F: $\mathbf{P(E \text{ or } F) = P(E) + P(F) - P(E \text{ and } F)}$ (subtract the overlap so it isn't counted twice; if the events are disjoint, $P(E \text{ and } F) = 0$ and this becomes the disjoint rule).

Example 12 (one card; $E = \text{king}$, $F = \text{diamond}$; overlap = king of diamonds) $P(\text{king or diamond}) = 4/52 + 13/52 - 1/52 = 16/52 = \mathbf{4/13 \approx 0.3077}$

Contingency table = a table that relates two categorical variables: a row variable and a column variable; each cell is a count.

Example 13 (marital status, millions, US 2021)

	Males	Females	Row total
Never married	49.5	43.2	92.7
Married	66.3	64.2	130.5
Widowed	3.5	11.5	15.0
Divorced	12.3	16.6	28.9
Separated	2.0	2.6	4.6
Column total	133.6	138.1	271.7

- (a) $P(\text{male}) = 133.6/271.7 \approx \mathbf{0.4917}$
- (b) $P(\text{widowed}) = 15.0/271.7 \approx \mathbf{0.0552}$
- (c) $P(\text{widowed or divorced}) = 15.0/271.7 + 28.9/271.7 = 43.9/271.7 \approx \mathbf{0.1616}$ (disjoint: a person has one status)
- (d) $P(\text{male or widowed}) = P(\text{male}) + P(\text{widowed}) - P(\text{male and widowed}) = 133.6/271.7 + 15.0/271.7 - 3.5/271.7 = 145.1/271.7 \approx \mathbf{0.5340}$

Complement of an Event

Complement of E, written E^c (or E' or "not E") = all outcomes in the sample space that are NOT in E.

Complement Rule: $\mathbf{P(E^c) = 1 - P(E)}$ (so $P(E) + P(E^c) = 1$). Illustration: a Venn diagram with the circle E inside the rectangle S; everything in the rectangle outside the circle is E^c .

Example 14: $P(\text{has played}) = 0.52 \rightarrow P(\text{has NOT played}) = 1 - 0.52 = \mathbf{0.48}$

4.3 Independence and the Multiplication Rule

Independent events = two events E and F are independent if the occurrence of E does NOT affect the probability of F (and vice versa). Otherwise they are **dependent**. - Coin flipped twice: first result doesn't

change the second → independent. - Die: $P(6) = 1/6$; but if told it was even, $P(6) = 1/3$ — the probability changed → "roll a 6" and "roll an even number" are dependent.

Example 15 - (a) Coin flip and die roll → **independent** (the coin can't affect the die). - (b) Bachelor's degree and earning over \$100,000 → **dependent** (having a degree changes the chance of a high income). - (c) Two randomly selected 24-year-old males' accidents → **independent** (one driver's accident doesn't affect the other's).

Multiplication Rule for Independent Events

Coin flipped twice: $S = \{HH, HT, TH, TT\}$ → $P(HH) = 1/4$. Separately: $P(H \text{ on 1st}) \times P(H \text{ on 2nd}) = 1/2 \times 1/2 = 1/4$ — same answer. If E and F are independent: **$P(E \text{ and } F) = P(E) \cdot P(F)$** (extends: $P(E \text{ and } F \text{ and } G) = P(E) \cdot P(F) \cdot P(G), \dots$).

Example 16 ($P(\text{a 24-year-old male survives the year}) = 0.9986$) - (a) Two survive: $0.9986 \times 0.9986 = 0.9986^2 \approx \mathbf{0.9972}$ - (b) Three survive: $0.9986^3 \approx \mathbf{0.9958}$ - (c) Twenty survive: $0.9986^{20} \approx \mathbf{0.9724}$

Computing At-Least Probabilities

"At least one" = 1 or more. **$P(\text{at least 1}) = 1 - P(\text{none})$** .

Example 17 (at least one of 1000 males dies; $P(\text{survive}) = 0.9986$) $P(\text{none die}) = P(\text{all 1000 survive}) = 0.9986^{1000} \approx 0.2464$ $P(\text{at least one dies}) = 1 - 0.9986^{1000} \approx 1 - 0.2464 = \mathbf{0.7536}$

4.4 Conditional Probability and the General Multiplication Rule

Conditional probability, $P(F | E)$ = the probability that F occurs GIVEN that E has already occurred (read "probability of F given E"). The "given" information shrinks the sample space to only the outcomes where E happened.

Example 18: $P(\text{six}) = 1/6$. Given the roll was even, the sample space is $\{2, 4, 6\}$ → $P(\text{six} | \text{even}) = \mathbf{1/3}$.

Conditional Probability Rule (marital example): $P(\text{widowed}) = 15.0/271.7 \approx 0.0552$. If we know the person is female, only the 138.1 million females are the sample space: $P(\text{widowed} | \text{female}) = 11.5/138.1 \approx \mathbf{0.0833}$.

Rule: for any two events E and F: **$P(F | E) = P(E \text{ and } F) \div P(E) = N(E \text{ and } F) \div N(E)$** (In a table: the cell count divided by the total of the row or column named after "given".)

Example 19 - (a) $P(\text{never married} | \text{male}) = 49.5/133.6 \approx \mathbf{0.3705}$ (divide by the male column total) - (b) $P(\text{male} | \text{never married}) = 49.5/92.7 \approx \mathbf{0.5340}$ (divide by the never-married row total)

General Multiplication Rule

$P(E \text{ and } F) = P(E) \cdot P(F | E)$ (works whether or not the events are independent).

Example 20: $P(\text{pulled over and ticket}) = P(\text{pulled over}) \times P(\text{ticket} | \text{pulled over}) = 0.8 \times 0.9 = \mathbf{0.72}$

Determining Independence

Rule of thumb: when sampling without replacement from a large population, if the sample is **less than 5% of the population**, treat the selections as independent.

E and F are independent if $P(F | E) = P(F)$ (equivalently $P(E | F) = P(E)$, or $P(E \text{ and } F) = P(E) \cdot P(F)$). If the probabilities are different, the events are dependent.

Example 21: $P(\text{widowed}) = 15.0/271.7 \approx 0.0552$, but $P(\text{widowed} | \text{female}) = 11.5/138.1 \approx 0.0833$. $0.0833 \neq 0.0552 \rightarrow$ **not independent (dependent)**: knowing the person is female raises the chance of being widowed.

4.5 Counting Techniques

General Counting Principle

Restaurant: 2 appetizers $\times$ 3 entrées $\times$ 2 desserts = **12** different meals (list: soup/salad $\times$ chicken/beef au jus/beef patty $\times$ ice cream/cheesecake).

Multiplication Rule of Counting: if a task is a sequence of choices with p options for the first, q for the second, r for the third, ..., the total number of ways is **$p \cdot q \cdot r \cdot \dots$**

Example 22 (3-letter airport codes, repeats allowed): $26 \times 26 \times 26 = 26^3 =$ **17,576**

Example 23 (chair, vice-chair, secretary from 14 people): $14 \times 13 \times 12 =$ **2,184** (no repeats: one fewer choice each time)

Factorial: $n! = n \cdot (n - 1) \cdot (n - 2) \cdot \dots \cdot 3 \cdot 2 \cdot 1$, and **$0! = 1$** . ($5! = 120$.)

Example 24 (visit 7 schools in some order): $7! = 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 =$ **5,040** routes

Permutations

Permutation = an ordered arrangement of r objects chosen from n distinct objects, no repeats. **ORDER MATTERS**. ${}_nP_r = n! \div (n - r)!$

Example 25: ${}_7P_5 = 7!/(7-5)! = 7!/2! = 5040/2 =$ **2,520**. ${}_5P_5 = 5!/0! = 120/1 =$ **120**. Excel: =PERMUT(7,5), =PERMUT(5,5).

Example 26 (10 runners; 1st, 2nd, 3rd): ${}_{10}P_3 = 10!/7! = 10 \times 9 \times 8 =$ **720**

Combinations

Combination = a selection of r objects from n distinct objects where **ORDER DOES NOT MATTER** (ABC is the same as BAC). ${}_nC_r = n! \div [r! (n - r)!]$

Example 27 (teams of 2 from Roger, Ken, Tom, Jay): list: RK, RT, RJ, KT, KJ, TJ $\rightarrow$ 6. Formula: ${}_4C_2 = 4!/(2! \cdot 2!) = 24/(2 \cdot 2) =$ **6**. Excel: =COMBIN(4,2).

Example 28 (SRS of size 4 from 20): ${}_{20}C_4 = 20!/(4! \cdot 16!) = (20 \cdot 19 \cdot 18 \cdot 17)/(4 \cdot 3 \cdot 2 \cdot 1) = 116,280/24 =$ **4,845**

Using Counting Methods in Probability

$P(E) = (\text{number of outcomes in } E) \div (\text{total number of outcomes})$, both found by counting.

Example 29 (dealt 4 cards; all 4 match = four of a kind) - (a) ${}_{52}C_4 = 52!/(4! \cdot 48!) = (52 \cdot 51 \cdot 50 \cdot 49)/24 =$ **270,725** possible hands - (b) One four-of-a-kind for each of the 13 ranks $\rightarrow$ **13** hands - (c) $P(\text{all 4 match}) = 13/270,725 \approx$ **0.00005** (≈ 0.0000480)

Which counting method? - Sequence of choices → Multiplication Rule of Counting ($p \cdot q \cdot r \dots$) - Choosing r of n objects, **order matters** → permutation ${}_nP_r$ - Choosing r of n objects, **order doesn't matter** → combination ${}_nC_r$

4.7 Determining Which Method to Use (the flow chart in words)

Single event? - Outcomes equally likely → classical: $P(E) = N(E)/N(S)$ - Not equally likely but data available → empirical: $P(E) \approx \text{frequency of } E / \text{number of trials}$ - No data → subjective probability

Two events — what word is in the question? - **"OR"** → disjoint? Yes: $P(E \text{ or } F) = P(E) + P(F)$. No: $P(E \text{ or } F) = P(E) + P(F) - P(E \text{ and } F)$ - **"AND"** → independent? Yes: $P(E \text{ and } F) = P(E) \cdot P(F)$. No: $P(E \text{ and } F) = P(E) \cdot P(F | E)$ - **"AT LEAST one"** → $P(\text{at least } 1) = 1 - P(\text{none})$ - **"GIVEN"** → conditional: $P(F | E) = P(E \text{ and } F)/P(E) = N(E \text{ and } F)/N(E)$ - **"NOT"** → complement: $P(\text{not } E) = 1 - P(E)$

Counting: sequence of choices → multiply ($p \cdot q \cdot r \dots$; tree diagram if choices depend on earlier ones). Selecting r of n → order matters: ${}_nP_r = n!/(n-r)!$; order doesn't matter: ${}_nC_r = n!/[r!(n-r)!]$. Excel: =PERMUT(n , r), =COMBIN(n , r).